

Andrew A. Fesenko

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EDUCATION

University of Florida Herbert Wertheim College of Engineering, Gainesville FL
Bachelor of Science in Computer Engineering

Expected Graduation - May 2027

EXPERIENCE

Palm Beach State College, Application Support Specialist

Aug 2025 - Present

- Develop and maintain backend APIs and webhook integrations to automate Student Services workflows across Admissions, Registration, Financial Aid, and Advising.
- Designed a Node.js service that authenticates Box using OAuth2, dynamically creates student folders, and uploads submission PDFs directly from JotForm webhooks.
- Configure and develop features within Modo Labs, including XModules and Modo APIs, to support mobile app and web portal functionality for students and staff.

Game-based Learning and Digital Experiences (GLaDE) Lab, Undergraduate Researcher

Feb 2025 - Present

- Built interactive VR simulations in Unity to teach engineering statics, increasing engagement and learning outcomes through immersive problem-solving.
- Designed 3D statics problems using Blender and SolidWorks, enabling virtual manipulation of forces and constraints in real time.
- Used C# and Python to generate dynamic problem variations with randomized inputs to reinforce core concepts.

Palm Beach State College, Web Development Intern

June 2024 - June 2025

- Worked on the redesign of the college website and mobile app, helping improve navigation and user experience based on feedback from students and staff.
- Built custom APIs to sync data between JotForm submissions and internal databases, streamlining workflows and reducing manual entry.
- Helped transition paper-based processes to secure online systems, speeding up processing and reducing errors across departments.

PROJECTS

Aegis, AI Powered PPE Access Control System (SwampHacks XI Winner)

- Built an automated PPE access control system that verifies compliance from a live camera feed before granting entry and records audit logs.
- Trained and deployed a YOLO based vision model and integrated Google Gemini to generate real time, user facing explanations of missing safety equipment.
- Implemented RFID and PIN authentication via ESP32, with a Flask vision service and a React admin dashboard using Socket.io and MongoDB for live analytics.

AI-Powered Interview Simulation Platform

- Built a job interview preparation platform using Next.js, Firebase, Tailwind, and Vapi AI with real-time voice-based interactions.
- Added feedback transcripts, Gemini-powered question generation, and a dashboard for tracking performance.
- Enabled real-time voice-based interviews with role-specific technical and behavioral questions

AI/NLP Series Analysis Project

- Built an AI/NLP system to analyze TV scripts, extract insights, and simulate character interactions.
- Developed a character network using Spacy's NER & NetworkX, visualizing relationships between characters.
- Implemented LLM-based classification, automating episode theme analysis with Hugging Face models.
- Created an interactive chatbot using Gradio & Transformers, enabling character-specific dialogues.

Arduino Self Balancing Cube

- Designed and built a self-balancing cube using ESP32, MPU6050 IMU, and Nidec 24H brushless motors.
- Implemented real-time PID control tuning, with Bluetooth-enabled calibration for dynamic balancing.
- Developed a real-time calibration system, allowing the cube to balance on an edge or vertex.

AI-Powered Basketball Game Analytics

- Built a computer vision system to analyze basketball games using YOLOv8, key point detection, and perspective transforms.
- Detected players and the ball in live footage, assigned team colors using zero-shot classification, and mapped positions to a top-down court view.
- Tracked ball possession, passes, and player movement automatically using deep learning, OpenCV, and NumPy.
- Used Hugging Face APIs to apply classification models and enhance tactical analysis from real-world video.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, Assembly (AVR, RISC-V), SQL, MATLAB, HTML/CSS, SystemVerilog, VHDL, SystemC
Tools & Platforms: Git, Node.js, Firebase, MongoDB, VS Code, JetBrains IDEs, RunPod, Adobe Creative Cloud, AutoCAD, SolidWorks, LTspice
FPGA and Digital Design: Intel Quartus Prime, ModelSim, Questa, AMD Vitis
Hardware & Embedded Systems: ESP32, ATxmega128A1U, UART, SPI, GPIO, USB, DAC, ADC, EBI, Atmel Studio

ADDITIONAL

Technical Interests: Hardware Systems, IoT, AI/ML, Cloud Computing, Embedded Systems, Real-Time Systems, Robotics, Automation, Systems Architecture